# Univariate Exploratory Data Analysis of NBA Dataset

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## 1 Data Validation

The dataset used in this analysis contains information on NBA games.

| Property | Value |
|---|---|
| Number of Rows | 70,905 |
| Number of Columns | 8 |
| Missing Values | 0 (0.00%) |

Table 1: Dataset dimensions and quality checks for the NBA dataset.

Overall, the dataset is complete with no missing values, and column names are consistent and standardized.

## 2 Identify Variable Types

Each column in the NBA dataset was classified according to its variable type (numeric, categorical, or date).

| Column Name | Type | Description |
|---|---|---|
| season | Integer | First year of season year of the game |
| date | Date | Game date (converted from string) |
| team1 | Categorical | Abbreviation for the first (home) team |
| team2 | Categorical | Abbreviation for the second team |
| result | Categorical | Game outcome for team1 (Win/Loss) |
| score1 | Numeric | Points scored by team1 |
| score2 | Numeric | Points scored by team2 |
| home_away | Categorical | Indicates if team1 played Home or Away |

Table 2: Classification of variables in the NBA dataset by type and description.

# 3 Descriptive Statistics

This section summarizes the statistical properties of both numeric and categorical variables in the NBA dataset. Numeric variables (`score1`, `score2`) are analyzed using measures of central tendency, dispersion, and distributional shape. Categorical variables (`result`, `home_away`, `team1`, `team2`) are analyzed using frequency counts and proportions.

## 3.1 Numeric Variables

| **Metric** | **Score1** | **Score2** |
|---|---|---|
| Number of Observations | 70905 | 70905 |
| Mean | 105.646 | 102.163 |
| Median | 105.0 | 102.0 |
| Mode | 106.0 | 103.0 |
| Standard Deviation | 14.396 | 14.069 |
| Variance | 207.254 | 197.937 |
| Skewness | 0.073 | 0.067 |
| Kurtosis | 0.118 | 0.144 |
| Minimum | 18.0 | 19.0 |
| Maximum | 184.0 | 186.0 |
| 1st Quartile (Q1) | 96.0 | 93.0 |
| 2nd Quartile (Q2, Median) | 105.0 | 102.0 |
| 3rd Quartile (Q3) | 115.0 | 111.0 |
| Interquartile Range (IQR) | 19.0 | 18.0 |

Table 3: Descriptive statistics of numeric variables (`score1` and `score2`).

## 3.2 Categorical Variables

| **Category** | **Count** | **Proportion (%)** |
|---|---|---|
| Win (result = W) | 43379 | 61.18 % |
| Loss (result = L) | 27526 | 38.82 % |

Table 4: Counts and proportions for the variable `result`.

| **Category** | **Count** | **Proportion (%)** |
|---|---|---|
| Home | 70905 | 100 % |
| Away | 0 | 0 % |

Table 5: Counts and proportions for the variable `home_away`.

| **Team1** | **Count** | **Proportion (%)** |
|---|---|---|
| BOS | 3486 | 4.92 |
| NYK | 3183 | 4.49 |
| LAL | 2987 | 4.21 |
| DET | 2936 | 4.14 |
| PHI | 2649 | 3.74 |
| CHI | 2586 | 3.65 |
| PHX | 2465 | 3.48 |
| MIL | 2451 | 3.46 |
| ATL | 2451 | 3.46 |
| POR | 2357 | 3.32 |
| CLE | 2353 | 3.32 |
| HOU | 2335 | 3.29 |
| GSW | 2316 | 3.27 |
| SAS | 2174 | 3.07 |
| WAS | 2144 | 3.02 |
| DEN | 2136 | 3.01 |
| IND | 2112 | 2.98 |
| UTA | 1998 | 2.82 |
| DAL | 1930 | 2.72 |
| SEA | 1755 | 2.48 |
| LAC | 1716 | 2.42 |
| SAC | 1648 | 2.32 |
| MIA | 1636 | 2.31 |
| ORL | 1513 | 2.13 |
| MIN | 1485 | 2.09 |
| NJN | 1467 | 2.07 |
| CHA | 1424 | 2.01 |
| TOR | 1259 | 1.78 |
| MEM | 1019 | 1.44 |
| OKC | 757 | 1.07 |
| BAL | 702 | 0.99 |
| CIN | 656 | 0.93 |
| BKN | 551 | 0.78 |

| **Team1** | **Count** | **Proportion (%)** |
|---|---|---|
| SLH | 498 | 0.70 |
| NOP | 496 | 0.70 |
| SYN | 482 | 0.68 |
| MNL | 426 | 0.60 |
| PHA | 423 | 0.60 |
| KCK | 421 | 0.59 |
| NOH | 379 | 0.53 |
| SFW | 353 | 0.50 |
| BUF | 346 | 0.49 |
| FWP | 338 | 0.48 |
| RCH | 276 | 0.39 |
| SDC | 246 | 0.35 |
| VAN | 230 | 0.32 |
| NOJ | 205 | 0.29 |
| SDR | 166 | 0.23 |
| INO | 153 | 0.22 |
| MLH | 146 | 0.21 |
| KCO | 126 | 0.18 |
| NOK | 82 | 0.12 |
| TCB | 69 | 0.10 |
| CHP | 54 | 0.08 |
| WSC | 53 | 0.07 |
| CHZ | 50 | 0.07 |
| CAP | 44 | 0.06 |
| NYN | 41 | 0.06 |
| AND | 38 | 0.05 |
| SLB | 33 | 0.05 |
| SBS | 32 | 0.05 |
| WAT | 31 | 0.04 |
| CHS | 31 | 0.04 |

Table 6: Counts and proportions for team identifiers (`team1`)

| Team2 | Count | Proportion (%) |
|---|---|---|
| BOS | 3289 | 4.64 |
| NYK | 3245 | 4.58 |
| LAL | 2942 | 4.15 |
| DET | 2857 | 4.03 |
| PHI | 2753 | 3.88 |
| CHI | 2530 | 3.57 |
| MIL | 2474 | 3.49 |
| PHX | 2466 | 3.48 |
| ATL | 2430 | 3.43 |
| POR | 2362 | 3.33 |
| HOU | 2348 | 3.31 |
| CLE | 2339 | 3.30 |
| GSW | 2313 | 3.26 |
| SAS | 2180 | 3.07 |
| WAS | 2149 | 3.03 |
| DEN | 2133 | 3.01 |
| IND | 2122 | 2.99 |
| UTA | 1998 | 2.82 |
| DAL | 1946 | 2.74 |
| SEA | 1792 | 2.53 |
| LAC | 1718 | 2.42 |
| SAC | 1650 | 2.33 |
| MIA | 1631 | 2.30 |
| ORL | 1518 | 2.14 |
| MIN | 1488 | 2.10 |
| NJN | 1472 | 2.08 |
| CHA | 1436 | 2.03 |
| TOR | 1257 | 1.77 |
| MEM | 1021 | 1.44 |
| OKC | 751 | 1.06 |
| SYN | 629 | 0.89 |
| SLH | 623 | 0.88 |
| CIN | 574 | 0.81 |
| PHA | 564 | 0.80 |
| BKN | 550 | 0.78 |
| BAL | 508 | 0.72 |
| NOP | 496 | 0.70 |
| MNL | 450 | 0.63 |
| KCK | 425 | 0.60 |
| SFW | 424 | 0.60 |
| NOH | 379 | 0.53 |
| BUF | 332 | 0.47 |
| RCH | 319 | 0.45 |
| FWP | 267 | 0.38 |
| SDC | 246 | 0.35 |
| VAN | 230 | 0.32 |
| NOJ | 205 | 0.29 |
| SDR | 168 | 0.24 |
| MLH | 136 | 0.19 |
| INO | 129 | 0.18 |
| KCO | 126 | 0.18 |
| NOK | 82 | 0.12 |
| TCB | 66 | 0.09 |
| WSC | 52 | 0.07 |
| CAP | 45 | 0.06 |
| NYN | 41 | 0.06 |
| CHS | 40 | 0.06 |
| SLB | 35 | 0.05 |
| AND | 34 | 0.05 |
| SBS | 33 | 0.05 |
| WAT | 31 | 0.04 |
| CHZ | 30 | 0.04 |
| CHP | 26 | 0.04 |

Table 7: Counts and proportions for team identifiers (`team2`)

| Team | Count | Proportion (%) |
|---|---|---|
| BOS | 6775 | 4.78 |
| NYK | 6428 | 4.53 |
| LAL | 5929 | 4.18 |
| DET | 5793 | 4.09 |
| PHI | 5402 | 3.81 |
| CHI | 5116 | 3.61 |
| PHX | 4931 | 3.48 |
| MIL | 4925 | 3.47 |
| ATL | 4881 | 3.44 |
| POR | 4719 | 3.33 |
| CLE | 4692 | 3.31 |
| HOU | 4683 | 3.30 |
| GSW | 4629 | 3.26 |
| SAS | 4354 | 3.07 |
| WAS | 4293 | 3.03 |
| DEN | 4269 | 3.01 |
| IND | 4234 | 2.99 |
| UTA | 3996 | 2.82 |
| DAL | 3876 | 2.73 |
| SEA | 3547 | 2.50 |
| LAC | 3434 | 2.42 |
| SAC | 3298 | 2.33 |
| MIA | 3267 | 2.30 |
| ORL | 3031 | 2.14 |
| MIN | 2973 | 2.10 |
| NJN | 2939 | 2.07 |
| CHA | 2860 | 2.02 |
| TOR | 2516 | 1.77 |
| MEM | 2040 | 1.44 |
| OKC | 1508 | 1.06 |
| CIN | 1230 | 0.87 |
| BAL | 1210 | 0.85 |
| SLH | 1121 | 0.79 |
| SYN | 1111 | 0.78 |
| BKN | 1101 | 0.78 |
| NOP | 992 | 0.70 |
| PHA | 987 | 0.70 |
| MNL | 876 | 0.62 |
| KCK | 846 | 0.60 |
| SFW | 777 | 0.55 |
| NOH | 758 | 0.53 |
| BUF | 678 | 0.48 |
| FWP | 605 | 0.43 |
| RCH | 595 | 0.42 |
| SDC | 492 | 0.35 |
| VAN | 460 | 0.32 |
| NOJ | 410 | 0.29 |
| SDR | 334 | 0.24 |
| MLH | 282 | 0.20 |
| INO | 282 | 0.20 |
| KCO | 252 | 0.18 |
| NOK | 164 | 0.12 |
| TCB | 135 | 0.10 |
| WSC | 105 | 0.07 |
| CAP | 89 | 0.06 |
| NYN | 82 | 0.06 |
| CHP | 80 | 0.06 |
| CHZ | 80 | 0.06 |
| AND | 72 | 0.05 |
| CHS | 71 | 0.05 |
| SLB | 68 | 0.05 |
| SBS | 65 | 0.05 |
| WAT | 62 | 0.04 |

Table 8: Counts and proportions for team identifiers (occurrences in `team1` and `team2` combined)

# 4 Visualizations

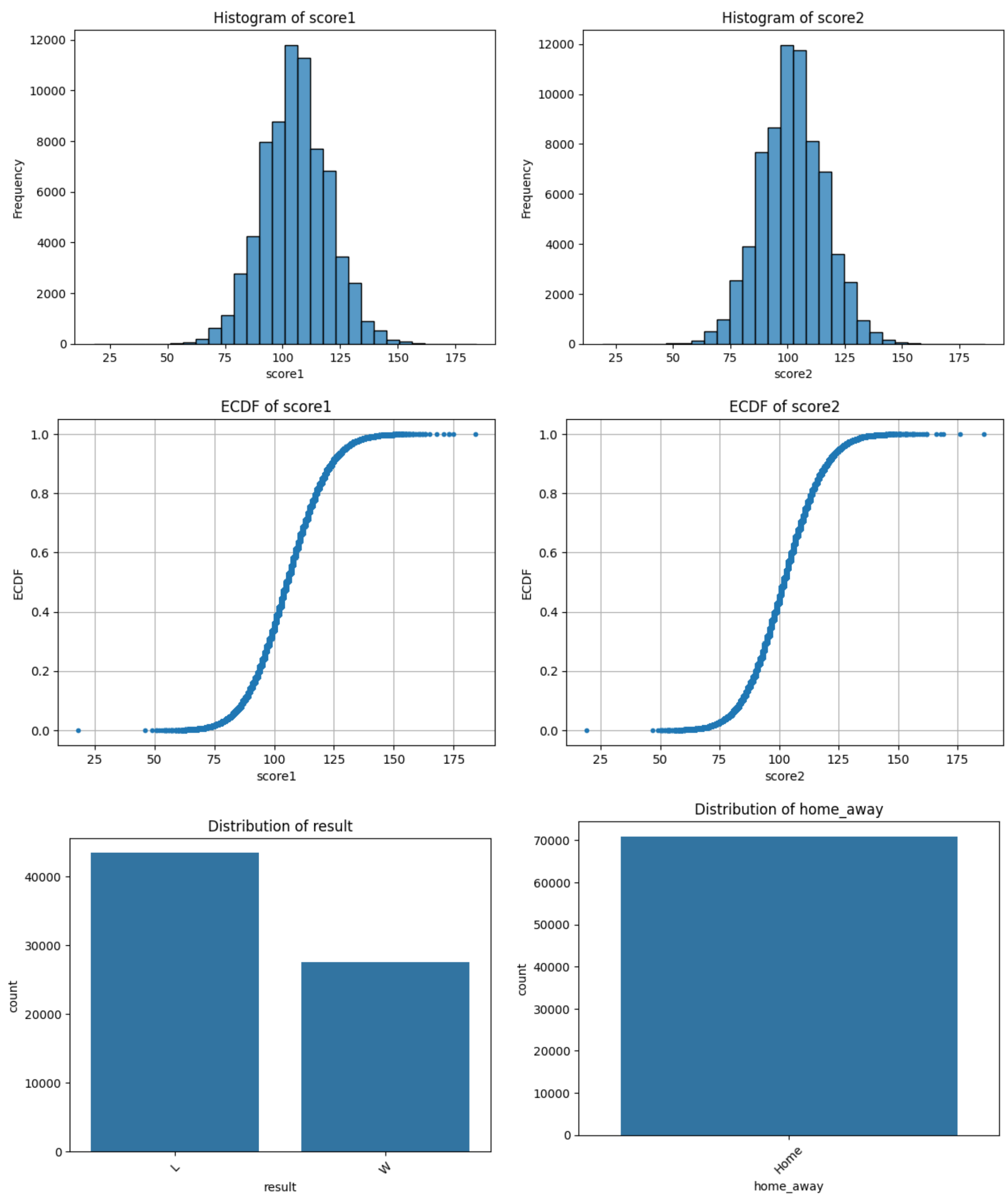


Figure 1: Descriptive visualizations of NBA game statistics: (a) Histogram of Score1, (b) Histogram of Score2, (c) ECDF of Score1, (d) ECDF of Score2, (e) Distribution of game results, (f) Home vs Away games.

# 5 Discussion

Table 1 shows the dataset contains 70,905 observations with 8 rows of data for each, and no missing values. In Table 3, we see that the numeric variables (`score1`, `score2`) have moderately symmetric descriptive statistics. The variability is moderate, with standard deviations at about 14 points, but `score1` is on average slightly higher than `score2`. The skewness values are close to zero, which implies that the data is not lopsided. The kurtosis values are slightly positive, which implies a distribution of data that is slightly more peaked than a normal distribution, which means a slightly higher concentration of values near the center of the distribution and fewer values at the tails. This interpretation of the kurtosis is reflected in the histograms for `score1` and `score2` shown in Figure 1.

In Table 4, the `result` variable shows evidence of a home-court advantage, with home teams winning 61.18% of games. Team frequencies represented in Table 6, Table 7, and Table 8, reflect the fact that the dataset spans 76 NBA seasons: long-standing franchises such as the Boston Celtics (BOS), New York Knicks (NYK) and Los Angeles Lakers (LAL) appear more often, while teams with short histories such as the Waterloo Hawks (WAT), Sheboygan Redskins (SBS) and St. Louis Bombers (SLB) appear at low frequencies.

In Figure 1, the empirical cumulative distribution functions (ECDFs) for `score1` and `score2` both produce a smooth, nearly linear progression between roughly 80 and 120 points. This reflects the fact that the majority of NBA game scores fall within this range. About 25% of scores are below 96 points and about 75% of scores are below 115 points, which is consistent with the quartiles reported in Table 3. The graphs of the functions also show the compactness of the distributions, with around 90% of all game scores falling between 85 and 120 points, and very small tails of extreme low and high outcomes. This reflects the fact that the rules and typical pace of play in NBA games result in natural boundaries on the distribution of points. Additionally, the median scores of 105 (`team1`) and 103 (`team2`), and the minimum scores of 18 (`team1`) and 19 (`team2`) show that even the lowest-scoring teams achieve meaningful point totals, and most score a significant amount. This could potentially be used to suggest that the NBA teams are generally at similar performance levels (every team is able to put up at least some fight against their opponent).

# 6 References

Data source: NBA dataset (nba_data.csv). This dataset contains data scraped and cleaned from https://www.basketball-reference.com/leagues on September 8, 2025. Data source used for EDA on September 13, 2025.